SOLAR ENERGY ESTIMATION FOR MINIMIZING ENERGY STORAGE REQUIREMENTS IN PV POWER PLANTS

A PROJECT REPORT

Submitted by

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DEEPAK M	**(620821205011)**

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

in

INFORMATION TECHNOLOGY

GNANAMANI COLLEGE OF TECHNOLOGY

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BONAFIDE CERTIFICATE

Certified that this project report "**SOLAR ENERGY ESTIMATION FOR MINIMIZING ENERGY STORAGE REQUIREMENTS IN PV POWER PLANTS"** is the bonafied work of **"ANANDHAKUMAR.S (620821205003), ARULSURIYAN.R (620821205004),DEEPAK.M (620821205011)** and **BHUVANESHWARWN.V (6208201205008)"** who carried out the project work under my supervisor.

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Submitted for the End Semester Project Work Viva-voce Examination held on_____________

INTERNAL EXAMINER

EXTERNAL EXAMINER

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VISION

Emerging as a technical institution of high standard and excellence to produce quality Engineers, Researchers, Administrators and Entrepreneurs with ethical and moral values to contribute the sustainable development of the society.

MISSION

We facilitate our students

- To have in-depth domain knowledge with analytical and practical skills in cutting edge technologies by imparting quality technical education.
- To be industry ready and multi-skilled personalities to transfer technology to industries and rural areas by creating interests among students in Research and Development and Entrepreneurship.

DEPARTMENT OF INFORMATION TECHNOLOGY

VISION

- To be the department that imparts professional computing training and makes competent engineer to work in the emerging areas of information technology field.

MISSION

- To prepare competent engineers and adapt to the dynamic needs of industries.
- To pave way to the enrichment of knowledge and skills using latest technologies in the diverse domain of information technology.
- Inculcate strong ethical values and professionalism to serve society while concurrently updating knowledge and skills through life-long learning.

ABSTRACT

Renewable Energy Sources (RES) are growing quickly all over the world thanks to both environmental and geopolitical concerns. However, the intermittent and stochastic production obtained from some of these RES poses technical and economic challenges when integrated on a large scale due to the introduction of significant uncertainties into the operation and planning of the power systems. This is especially important for wind or photovoltaic (PV) power technologies. Wind power is currently more widespread, globally speaking, and the aforementioned problem has thus already been tackled in countries such as Denmark. Although a traditional solution to the intermittency of wind generation has been primarily based on improving grid interconnection, as wind penetration becomes more and more important new solutions have had to be proposed. The abstract could describe the solar system as a gravitationally bound system consisting of the Sun, eight planets, numerous moons, asteroids, and comets, all orbiting the Sun within the Milky Way galaxy. It could also briefly mention the system's formation and key features, such as the difference between terrestrial and gas giant planets. The abstract should be concise and informative, suitable for a general audience interested in astronomy. The solar system is a gravitationally bound system centered around the Sun, our star.

It includes eight planets (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, Neptune), numerous moons, and a diverse population of asteroids, comets, and dwarf planets. The solar system is located within the Milky Way galaxy and formed from a collapsing nebula. Key features include the asteroid belt, the Kuiper belt, and the Oort cloud. Understanding the solar system provides insights into planetary formation, the dynamics of celestial bodies, and the search for life beyond Earth. The solar system was formed as the result of the collapse of a cloud of pre-existing interstellar gas and dust. We should therefore expect a close compositional relationship between the solar system and the interstellar material from which it formed. Our solar system includes the Sun, eight planets, five officially named dwarf planets, hundreds of moons, and thousands of asteroids and comets. Our solar system is located in the Milky Way, a barred spiral galaxy with two major arms, and two minor arms. The solar system is a gravitationally bound system consisting of the Sun, its eight planets (Mercury, Venus, Earth, Mars, Jupiter, Saturn, Uranus, and Neptune), numerous moons, asteroids, comets, and other minor bodies. It's located in the Milky Way Galaxy.

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LIST OF ABBERIVATION

GLCM	Gray level co-occurrence matrix
GLRLM	Gray level run length matrix
GLSZM	Gray level size zone matrix
NGTDM	Neighbouring Gray tone difference matrix
GLDM	Gray level dependence matrix
DCNN	Deep Convolutional Neural Network
CNN	Convolutional Neural Network
LRN	local response normalization
ROC Curve	Receiver Operating Characteristic Curve
AUC	Area Under the ROC Curve
Mean	Average Precision (mAP)

CHAPTER 1
INTRODUCTION

1.1. GENERAL INTRODUCTION

Solar panels are devices that convert sunlight into electricity using a technology called photovoltaics. They are made up of many solar cells, usually made from silicon, which capture sunlight and turn it into usable electrical energy.Solar panels are an important part of renewable energy systems. They provide a clean, eco-friendly source of power without producing harmful emissions. Commonly used on rooftops, buildings, and in solar farms, they help reduce dependence on fossil fuels and lower electricity bills. As technology improves, solar panels are becoming more efficient and affordable, making them a key solution in the fight against climate change and in promoting sustainable energy. The solar system is a group of celestial objects that orbit around the Sun, which is a star at the center. It includes eight major planets, their moons, dwarf planets, asteroids, comets, and other space objects. Each planet moves around the Sun in a path called an orbit due to the Sun’s strong gravitational pull.The solar system formed about 4.6 billion years ago from a cloud of gas and dust. Over time, gravity pulled most of the mass into the center to form the Sun, while the rest formed the planets and other objects.

1.2. IMPORTANCE OF THE STUDY:

Understanding Earth's Place in the Universe:

Studying the solar system helps us learn where Earth fits in the vast universe. It gives us a broader perspective on how unique our planet is and how conditions came together to support life.

Learning About Planetary Formation:

By studying other planets and celestial bodies scientists can understand how the solar system formed around 4.6 billion years ago. This also helps explain how planets evolve and why they differ in size, composition, and atmosphere.

Clues to the Origin of Life:

Comparing conditions on Earth with those on Mars, Europa, or Enceladus can provide clues about whether life might exist elsewhere. It also helps us understand how life began on Earth.

Preparing for Space Exploration and Colonization:

Knowing more about other planets, especially Mars and the Moon, is vital for future human space missions. This includes learning about their environments, resources, and the challenges of living there.

Predicting and Preventing Threats:

The study of asteroids, comets, and space weather (like solar flares) helps us detect and respond to potential threats that could impact Earth, such as asteroid collisions or geomagnetic storms.

Technological Advancements:

Space research often leads to new technologies that benefit life on Earth, including better satellite systems, communication tools, medical devices, and materials.

1.3.PROBLEM STATEMENT:

A common problem statement for a solar system project focuses on the challenge of maximizing energy output, especially in locations like Ellar, Tamil Nadu, where seasonal variations and dust accumulation can affect solar panel efficiency. This systems that can adapt to changing solar angles and ensure optimal performance despite environmental factors. The intensity and angle of sunlight vary throughout the day and year. A static solar panel installation may not always be optimally aligned to capture the maximum amount of sunlight. Manual cleaning of solar panels is labor-intensive and can be costly, especially for large-scale installations. One of the most common problems with solar panels is that they can get covered with mud, dirt, and debris. Also, over time, bird droppings and pollen also accumulate on the panels, reducing their efficiency.

A common problem statement for solar systems revolves around balancing the benefits of solar energy with its limitations, particularly the intermittency of sunlight and the challenges of efficient energy storage.

1.4.AIM AND OBJECTIVE:

Aim:

The main aim of solar panel technology is to convert sunlight into usable electricity, offering a renewable and sustainable energy alternative. This can lead to reduced pollution, lower energy costs, increased energy independence, and a move towards a more sustainable and eco-friendly future. In India, the National Solar Mission aims to establish the country as a global leader in solar energy by promoting its use and diffusion.

This process aims to provide clean and renewable energy, reducing dependence on fossil fuels and promoting energy independence. Solar systems also play a crucial role in reducing carbon footprints, lowering electricity bills, and improving energy security.

Objectives:

- Solar Energy Devices have the potential to unlock the vast segments of the market that were previously without electricity due to high upfront costs of the systems themselves.
- The development of affordable, inexhaustible clean solar energy technologies will have huge longer term benefits.
- It will increase country's energy security through reliance on an indigenous, inexhaustible and mostly import- independent resource, enhance sustainability, reduce pollution, lower the cost of mitigating global warming and keep fossil fuel prices lower than otherwise.
- . Additionally, studying the solar system helps us understand the environment it resides in and search for evidence of past and present life on other worlds.

The main objectives of the study are as below:

1. To study the need and awareness of solar energy devices in rural & urban areas.
2. To analyze the Position of Solar Energy Devices in Market.
3. To define the market in which the product or brand will compete.
4. To give suggestion for the better utilizationfor the market by solar companies.
5. The primary aim of a solar home system is to provide households with a reliable and sustainable source of electricity.
6. Understand the structure of the solar system.
7. Learn planetary characteristics.
8. Explore gravitational relationships.
9. Understand scale and distances.

1.5. Scope and Limitation of the Study:

The scope of study for the solar system encompasses understanding its formation, structure, composition, and dynamics, as well as its role in planetary systems and the possibility of life beyond Earth. This includes studying planets, moons, asteroids, comets, meteorites, and the heliosphere. It also involves comparative planetology, planetary atmospheres and interiors, and the search for exoplanets. Studying the early solar system, including the protoplanetary disk and the processes that led to planet formation. Analyzing the compositions of meteorites and other small bodies to understand the building blocks of planets.

The scope of solar energy refers to its broad applications and potential for energy generation, while limitations highlight challenges like intermittency, storage, and efficiency. Solar energy is a renewable resource, meaning it's replenished by the sun, and it can be used for various purposes including electricity generation, heating, and cooling. However, its use is limited by factors like the availability of sunlight, the need for large areas for solar farms, and the cost of storage solutions.

This can lead to reduced pollution, lower energy costs, increased energy independence, and a move towards a more sustainable and eco-friendly future. In India, the National Solar Mission aims to establish the country as a global leader in solar energy by promoting its use and diffusion. This also helps explain how planets evolve and why they differ in size, composition, and atmosphere.

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CHAPTER 2

LITERATURE REVIEW

2.1. Hybrid Photovoltaic/Thermal Solar Systems:

We present test results on hybrid solar systems, consisting of photovoltaic modules and thermal collectors (hybrid PV/T systems). The solar radiation increases the temperature of PV modules, resulting in a drop of their electrical efficiency. By proper circulation of a fluid with low inlet temperature, heat is extracted from the PV modules keeping the electrical efficiency at satisfactory values. The extracted thermal energy can be used in several ways, increasing the total energy output of the system. Hybrid PV/T systems can be applied mainly in buildings for the production of electricity and heat and are suitable for PV applications under high values of solar radiation and ambient temperature.

2.2. Solar Radiation Model for a Power System:

The dwindling number of conventional power resources and its environmental impact has motivated a transition to renewable energy sources, such as solar power. Evaluating the reliability of solar power integration into power networks can help decision makers gauge the feasibility of their solar power projects. However, the stochastic and non-stationary nature of solar radiation is difficult to be modeled and can even hinder an accurate evaluation of reliability. A good solar model for accurately assessing solar-powerintegrated systems should be able to retain the original statistical properties of the sampled solar radiation data.

2.3. Solar Energy for Future World:

World's energy demand is growing fast because of population explosion and technological advancements. It is therefore important to go for reliable, cost effective and everlasting renewable energy source for energy demand arising in future. Solar energy, among other renewable sources of energy, is a promising and freely available energy source for managing long term issues in energy crisis. Solar industry is developing steadily all over the world because of the high demand for energy while major energy source, fossil fuel, is limited and other sources are expensive. It has become a tool to develop economic status of developing countries and to sustain the lives of many underprivileged people as it is now cost effective after a long aggressive researches done to expedite its development.

2.4. Design and Simulation of Solar PV System:

The demand of the electrical power is increasing per day which is supplied by fossil fuels resulting into huge carbon emissions in the atmosphere, which leads the electrical engineers to generate the power by using the renewable energy sources. This paper is aimed at simulation and development of Solar PV system which is able to fulfil the power demand in the isolated locations or in standalone condition. The system consists of various components like PV solar panel, DC-DC converter (Step up converter) and two level inverter connected to load. The controlling of input loop of the solar PV system is shown with the help of PI controller for maintaining the dc link constant irrespective of changes in the input side and output side parameters resulting into the constant inverter output. The simulations are performed in the MATLAB SIMULINK software.

2.5. MPPT Method for Low-Power Solar Energy Harvesting:

This paper describes a new maximum-power-pointtracking (MPPT) method focused on low-power (< 1 W) photovoltaic (PV) panels. The static and dynamic performance is theoretically analyzed, and design criteria are provided. A prototype was implemented with a 500-mW PV panel, a commercial boost converter, and low-power components for the MPPT controller. Laboratory measurements were performed to assess the effectiveness of the proposed method. Tracking efficiency was higher than 99.6%. The overall efficiency was higher than 92% for a PV panel power higher than 100 mW.

2.6. Performance Under Temperature and Solar Radiation:

This paper demonstrates a mathematical representation of Photovoltaic (PV) solar cells and hence panels performance. One-diode solar cell model is implemented to simulate the cell and extract the performance indications. The tested PV modules are BP Solar (60 Watt) and Synthesis Power (50 Watts), which are operating in a PV generation system in the University of Anbar - Iraq, College of Applied Sciences. The math model demonstrates Power versus Voltage (P-V) characteristic curves to depict and study various parameters with affecting variations in the PV array performance.

CHAPTER 3
METHODOLOGY

3.1. EXISTING SYSTEM:

There are different type of models in Existing system:

- Astronomical model
- Solar energy systems
- Simulation or software systems
- Educational models
- Infrastructure and satellites

Astronomical model:

The currently accepted model is the heliocentric model,

Where,

The Sun is at the center of the solar system. Planets orbit the Sun in elliptical (oval-shaped) paths, as described by Kepler’s laws of planetary motion. Gravity, as explained by Newton's law of universal gravitation and refined by Einstein's general relativity, governs these motions.

Key Features:

- **The Sun** – A G-type main-sequence star that contains over 99.8% of the solar system’s mass.
- **Planets** – Eight major planets (in order from the Sun):
  - Terrestrial: Mercury, Venus, Earth, Mars
  - Gas/ice giants: Jupiter, Saturn, Uranus, Neptune
- **Dwarf Planets** – Pluto, Eris, Haumea, Make and Ceres (in the asteroid belt).
- **Moons** – Over 200 known natural satellites orbiting planets.
- **Small Bodies** – Asteroids (mainly in the asteroid belt), comets (from the Kuiper Belt and Oort Cloud), and meteoroids.
- **The Kuiper Belt** – A region beyond Neptune containing many icy bodies.

1. **The Oort Cloud :**

   A hypothesized distant shell of icy objects surrounding the solar system, not yet directly observed.

2. **Models of the Past:**

   Geocentric model (Earth-centered), proposed by Ptolemy,dominated until the 16th century. Heliocentric model, proposed by Copernicus, confirmed and refined by Galileo, Kepler, and Newton.

Advantages:

- Renewable Energy Source
- Environmentally Friendly
- Reduces Electricity Bills
- Low Operating Costs
- Technology Maturity
- Scalability
- Energy Independence
- Government Incentives
- Long Lifespan
- Increases Property Value

Disadvantages:

- High Initial Cost
- Energy Storage is Expensive
- Requires Space
- Efficiency Limitations
- Aesthetic Concerns
- Installation Constraint
- Energy Production Varies by Location
- Environmental Impact of Manufacturing

3.2.PROPOSED SYSTEM:

There are many proposed system methods,

- **Satellite imagery:**

  Utilizes data from satellites to capture wide-area views of cloud cover, albedo, and other factors affecting solar irradiance.

- **Geospatial data:**

  This might include digital elevation models (DEMs), land use maps, and 3D models of buildings for precise solar radiation calculations

- **Historical data:**

  Long-term records of solar radiation, weather patterns, and energy consumption can be used to train and refine models.

1.Advanced Algorithms and Models:

- **Machine learning:**

  CNNs and LSTMs can learn complex patterns from input data to predict solar radiation and power output.

- **Hybrid neural networks:**

  Combining different neural network architectures (e.g., CNN for feature extraction and LSTM for temporal relationships) can improve prediction accuracy.

- **Optimization algorithms:**

  Differential evolution (DE), particle swarm optimization (PSO), and genetic algorithms can be used to optimize system configurations (e.g., panel orientation, system size) for maximum energy generation.

- **Geostatistical methods:**

  Spatial interpolation and clustering techniques can be applied to map solar radiation potential across regions.

2. System Design and Implementation:

- **Data pre processing and feature extraction:**

  Input data needs to be cleaned, formatted, and transformed into features suitable for model training.

- **Model training and validation:**

  Selected models are trained using historical data and validated against independent data sets to assess performance.

- **Output and visualization:**

  The system can generate predictions of solar radiation, power output, and other relevant parameters. Results can be visualized on maps or charts.

- **Integration with other systems:**

  The solar energy estimation system can be integrated with other energy management systems or smart grids to optimize energy use.

3.Specific Applications:

- **Rooftop solar potential assessment:**

  Determining the solar energy potential of building rooftops for solar panel installations.

- **Regional solar resource mapping:**

  Creating maps of solar radiation potential across a region to aid in planning solar power projects.

- **Solar power forecasting:**

  Predicting future solar radiation levels for grid integration and energy management.

- **Bifacial PV system modelling:**

  Estimating solar irradiance on both sides of bifacial panels for more accurate power output predictions.

4.Microgrid optimization:

Using solar energy estimation as part of a larger optimization process for microgrids. Microgrid optimization in solar energy involves maximizing the efficiency and effectiveness of local energy systems powered by solar photovoltaic (PV) and other renewable sources. This optimization aims to reduce costs, improve reliability, and enhance energy management within the microgrid. It involves strategies like demand response, energy storage management, and efficient resource allocation.

Advantages :

- Enhanced efficiency through AI-powered camera traps for automated wildlife monitoring.
- Real-time data collection provides immediate insights into wildlife movements.
- Accurate species identification using the WildNet model based on transfer learning.
- Proactive early warning system mitigates risks with timely SMS notifications and alarms.
- Contribution to reduced wildlife-vehicle collisions, enhancing overall road safety.
- User-friendly interfaces for easy system management and active user participation.
- Adaptability and scalability for application in diverse regions with wildlife crossings.
- Estimating solar irradiance on both sides of bifacial panels for more accurate power output predictions.

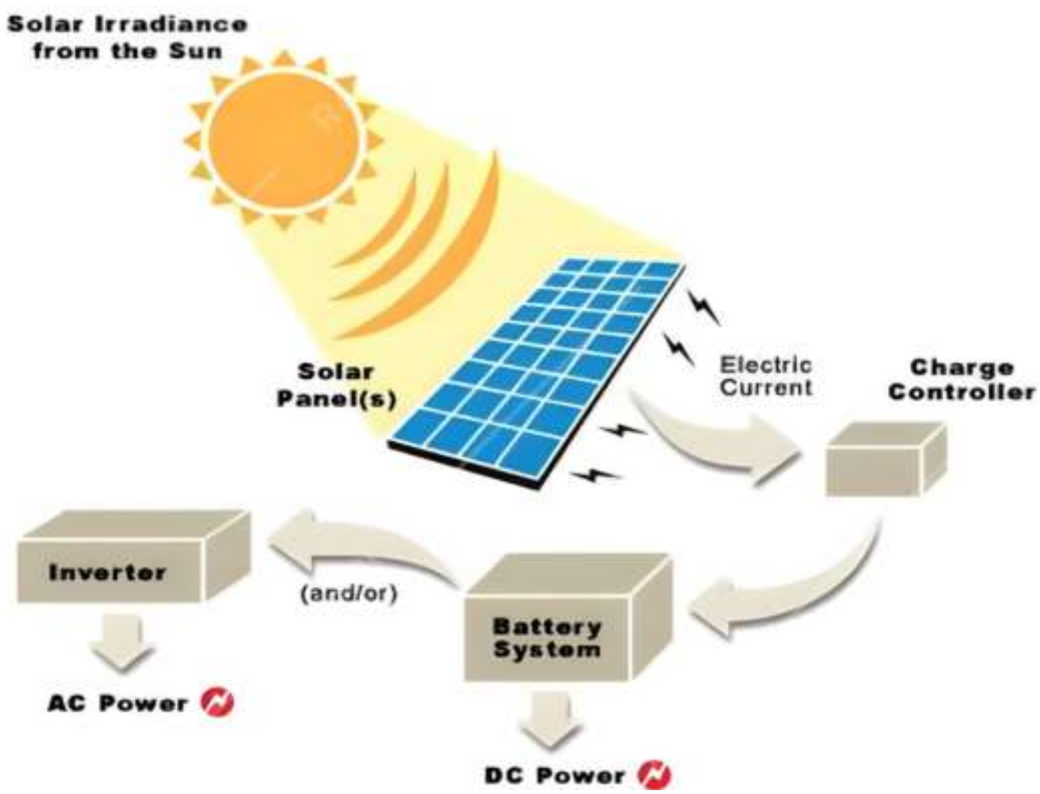
Disadvantages:

- Even though computers were created to make human life easier, they are currently only utilized for leisure.
- A power supply is necessary for the desktop computer to turn on and perform fully.
- Spending long hours on computers can lead to several health issues, including eye strain, headaches, and digital eye syndrome, often due to screen glare and blue light exposure.
- Computers face numerous security threats like hacking, phishing scams, and malware, putting users' personal and financial data at risk.
- Disposing of computers and electronic devices adds to global electronic waste, releasing toxic substances like mercury, lead, and cadmium into the environment.
- The manufacturing process of solar panels involves the use of hazardous materials and can generate waste, contributing to environmental pollution. Additionally, disposal of old panels can be problematic and environmentally damaged.

CHAPTER 4

WORKING OF SYSTEM

4.1. SYSTEM ARCHITECTURE:



1.Solar irradiance from the sun:

Solar irradiance from the sun refers to the power per unit area received from the sun in the form of electromagnetic radiation, measured in watts per square meter (W/m^2). It’s a key concept in solar energy and climate studies, and it directly impacts how much energy a solar panel can produce.

Types of Solar Irradiance:

Global Horizontal Irradiance (GHI): Total solar radiation received per unit area by a surface horizontal to the ground.

Direct Normal Irradiance (DNI): Solar radiation received directly from the sun, without scattering.

Diffuse Horizontal Irradiance (DHI): Solar radiation received from the sky (excluding direct sunlight), scattered by molecules and particles.

Influencing Factors:

- Time of day (highest at noon).
- Season (higher in summer, lower in winter).
- Latitude and altitude (closer to equator = higher irradiance).
- Weather conditions (clouds, haze, pollution reduce irradiance).
- Angle and orientation of the surface receiving sunlight.

2 Solar Panels:

Solar panels are devices that convert sunlight into electricity using the photovoltaic (PV) effect. They are a cornerstone of renewable energy systems, providing a clean, sustainable alternative to fossil fuels.

Main Components of a Solar Panel System

- **Solar Panels (PV Modules):** Convert sunlight into electricity.
- **Inverter:** Converts DC to AC electricity.
- **Mounting System:** Secures panels to a roof or ground structure.
- **Battery Storage (optional):** Stores excess power for later use.
- **Charge Controller (in off-grid systems):** Regulates power going to the batteries.
- **Monitoring System:** Tracks system performance and energy production.

Benefits:

- Reduces electricity bills.
- Decreases carbon footprint.
- Low maintenance and long lifespan (25–30 years).
- Increases property value.

3.Charge Controller:

A charge controller is a critical component in solar power systems especially off-grid and hybrid setups that regulates the voltage and current coming from the solar panels going into the battery. Its main job is to prevent overcharging, over-discharging, and battery damage.

Main Functions of a Charge Controller:

- **Prevents Overcharging**

Stops the battery from receiving too much voltage or current, which can shorten its lifespan or cause overheating.

- **Prevents Over-Discharging**

Disconnects the battery from the load when voltage drops too low, protecting battery health.

- **Voltage Regulation**

Ensures the solar panel voltage is adjusted to match battery requirements.

- **Temperature Compensation**

Some controllers adjust charge parameters based on ambient or battery temperature.

- **Status Monitoring and Safety**

Displays battery level, charging status, and may include safety features like short-circuit protection.

4.Battery System:

A battery system in a solar power setup stores excess energy generated by solar panels for later use especially useful during the night, cloudy days, or power outages. It's a key part of off-grid and hybrid solar systems, and increasingly common in grid-tied systems with backup.

Displays battery level, charging status, and may include safety features like short-circuit protection.

Key Functions of a Solar Battery System:

- **Energy Storage –** Stores surplus solar energy during the day.
- **Backup Power –** Provides electricity when the sun isn't shining or during outages.
- **Load Shifting –** Enables use of stored energy during peak electricity rate hours.
- **Grid Independence –** Reduces reliance on the utility grid.
- **Charge Controller –** Regulates the energy flow from solar panels to batteries.

5.Inverter:

An inverter is a vital component in a solar power system that converts the direct current (DC) electricity produced by solar panels into alternating current (AC) electricity, which is the standard form of electricity used by homes, businesses, and the power grid.

Types of Solar Inverters:

1. String Inverters
2. Microinverters
3. Power Optimizers

4. Hybrid Inverters

Key Features:

- Efficiency (look for 95% or higher)
- Maximum power output
- Battery compatibility (for hybrid/off-grid setups)
- Monitoring capabilities (Wi-Fi, app-based)
- Safety features (anti-islanding, surge protection)
- Low maintenance and long lifespan (25–30 years).
- Increases property value.
- Reduces electricity bills.
- Decreases carbon footprint.

4.2. IoT Configuration:

- **Microcontroller / Gateway** (e.g., Raspberry Pi, ESP32, Arduino + GSM/Wi-Fi/LTE)
- **Sensors:**
    - Voltage & Current Sensors (e.g., INA219)
    - Temperature Sensors (e.g., DS18B20)
    - Irradiance Sensor
    - Battery Level Sensors.

Microcontroller / Gateway:

An IoT gateway is a physical device or virtual platform that connects sensors, IoT modules, and smart devices to the cloud. Gateways serve as a wireless access portal to give IoT devices access to the Internet. In the context of the Internet of Things (IoT), an MCU, or microcontroller unit, is a small, self-contained computer on a single chip. It's used to control specific tasks within an embedded system, often in IoT devices. MCUs are chosen for IoT applications due to their ability to provide sufficient computing power while keeping costs, complexity, and energy consumption low.

Processing Power:

MCUs have a processor core that can handle data from sensors, execute pre-programmed algorithms, and make decisions.

Memory:

They include both RAM (Random Access Memory) and non-volatile memory types (like flash memory) for storing data and instructions.

Input/Output (I/O):

MCUs have I/O pins that allow them to interact with sensors, actuators, and other peripherals.

Communication:

MCUs can facilitate communication with other IoT devices and cloud services via various protocols like Wi-Fi, Bluetooth, and Zigbee.

Low Power:

Many IoT applications require battery-powered devices, making low-power MCUs a suitable choice.

Embedded Systems:

MCUs are ideal for embedded systems, where they control the operation of specific devices or tasks without a complex operating system.

Voltage & Current Sensors (e.g., INA219):

The INA219 senses across shunts on buses that can vary from 0 to 26 V. The device uses a single 3- to 5.5-V supply, drawing a maximum of 1 mA of supply current. The INA219 operates from –40°C to 125°C. MCUs are chosen for IoT applications due to their ability to provide sufficient computing power while keeping costs, complexity, and energy consumption low.

They include both RAM (Random Access Memory) and non-volatile memory types (like flash memory) for storing data and instructions.

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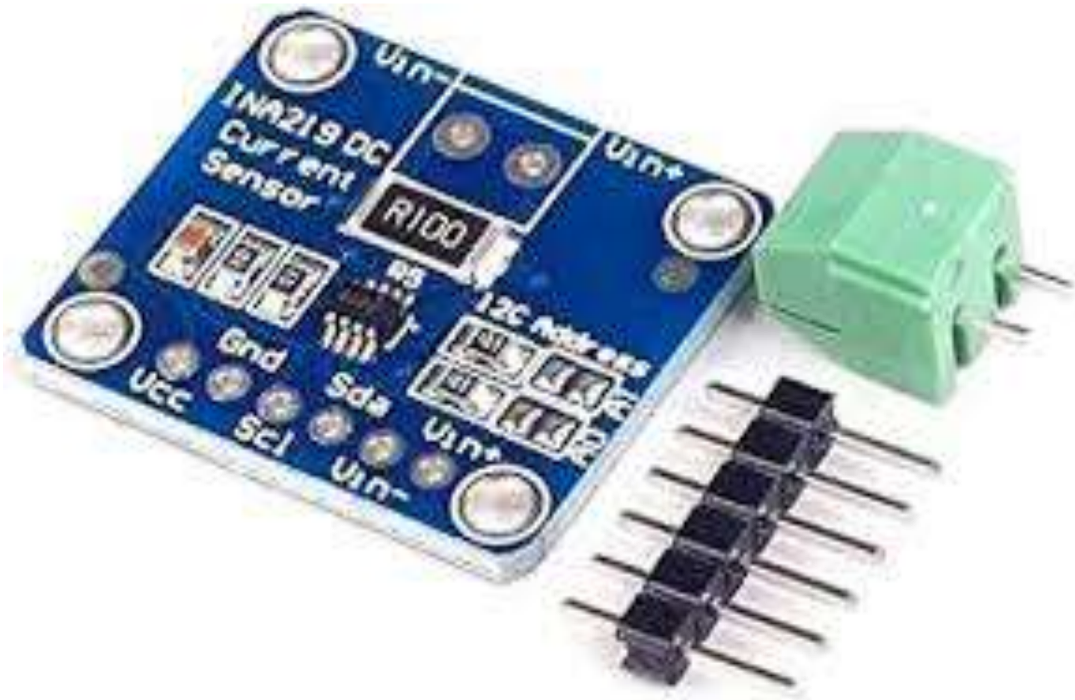
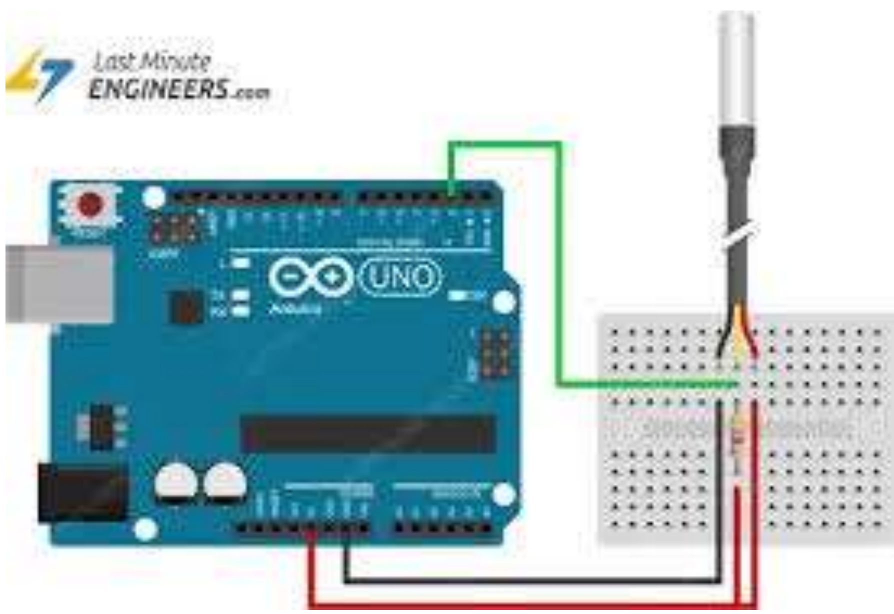


Temperature Sensors (e.g., DS18B20):

The DS18B20 is a digital temperature sensor that communicates using the 1-Wire protocol, requiring only one digital pin for communication. It offers a usable temperature range of -55 to 125°C, with an accuracy of ±0.5°C from -10°C to +85°C. Each sensor has a unique 64-bit ID, enabling multiple sensors to be connected to the same data line. MCUs are chosen for IoT applications due to their ability to provide sufficient computing power while keeping costs, complexity, and energy consumption low.

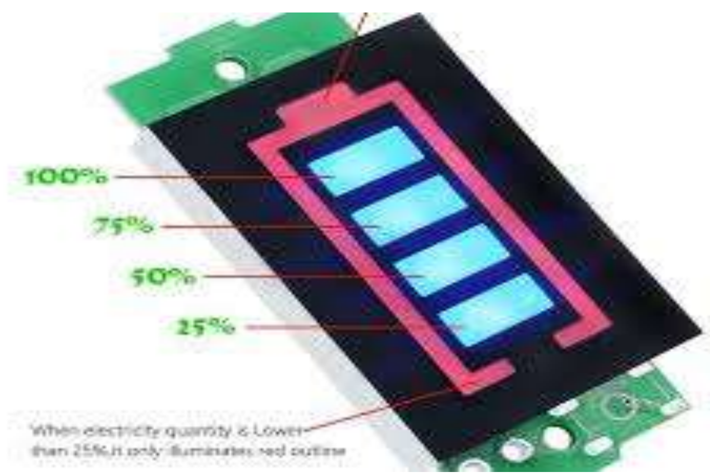


Irradiance Sensor:

An irradiance sensor, also known as a pyranometer, is a device that measures the amount of solar radiation received on a specific surface. It quantifies the global shortwave radiation, which includes both direct and diffuse sunlight, in watts per square meter. These sensors are commonly used in photovoltaic (PV) systems to monitor solar energy availability and optimize system performance**.**

Battery Level Sensors:

The electronic battery sensor (EBS) is attached to the negative terminal of a 12V lead-acid battery with the terminal clamp and connected to the vehicle's body by a screw-on ground cable. The EBS measures the current using a shunt and determines the battery's voltage and temperature.



CHAPTER 5
SYSTEM CONFIGURATION

5.1. HARDWARE REQUIREMENTS :

1. Solar Panels
2. Inverter
3. Battery Bank
4. Charge Controller
5. Mounting Structure
6. Wiring and Cables
7. Connectors and Junction Boxes
8. Protective Devices

1. Solar Panels (PV Modules):

- **Type**: Monocrystalline (high efficiency), Polycrystalline, Thin-film
- **Specs to Consider**:
    - Wattage (e.g., 330W, 450W)
    - Efficiency (%)
    - Temperature coefficient
    - Warranty (usually 25 years)

2. Inverter:

- **On-grid Systems**: String inverters or microinverters.
- **Off-grid/Hybrid Systems**: Hybrid or off-grid inverters.
- **Brands**: SMA, Fronius, Growatt, Huawei, Enphase (for microinverters)

3. Battery Bank (for off-grid or hybrid systems):

- **Types:**
    - Lead-acid (Flooded or VRLA)
    - Lithium-ion (LiFePO4 preferred)

- **Capacity: kWh based on backup requirements**
- **BMS (Battery Management System): Included with smart batteries**

4. Charge Controller (mainly in off-grid systems):

- **Type**: MPPT (preferred for efficiency), PWMRegulates voltage/current from panels to batteries

5. Mounting Structure:

- **Types**: Roof-mount, ground-mount, pole-mount
- **Material**: Galvanized steel, aluminum
- **Tilt Angle**: Adjusted based on location's latitude

6. Wiring and Cables:

- **DC Cables**: PV-rated cables (e.g., $4mm^2$, $6mm^2$)
- **AC Cables**: Based on inverter output
- **Earthing Wire**: For grounding

7. Connectors and Junction Boxes:

- **MC4 Connectors**: For connecting PV modules.
- **Combiner Boxes**: Combine outputs from multiple strings.
- **DCDB/ACDB**: DC/AC distribution boxes with surge protection and breakers.

8,Protective Devices:

- **Surge Protection Devices (SPD)**
- **Miniature Circuit Breakers (MCB)**
- **Fuses**: Rated for DC and AC circuits.
- **Isolators/Disconnects**: For system shutdown and maintenance.

9. Monitoring System:

- **Inverter Monitoring**: Usually built-in (via Wi-Fi, Ethernet, or GSM).
- **Advanced Options**: Data loggers or third-party smart energy meters.

10. Grounding and Lightning Protection:

- **Earthing Rods**: For all metallic components.
- **Lightning Arrestors**: Especially important in high-risk areas.

5.2. SOFTWARE REQUIREMENTS:

- Programming Languages
- IoT Cloud Platforms
- Libraries & Tools
- Other Tools

5.3. SOFTWARE DESCRIPTIONS:

Languages Used:

- **C/C++** – Standard for Arduino/ESP32 firmware development.
- **Micro Python** – For ESP boards with lightweight scripting.
- **Python** – Especially for Raspberry Pi or cloud-based processing.
- **JavaScript (Node.js)** – If integrating with web servers or dashboards.

IoT Cloud Platforms:

- **Thing Speak:**
    - Great for quick visualization and free usage.
    - Compatible with ESP boards, uses HTTP/MQTT.
- **Firebase:**
    - Real-time database for mobile/web apps.
    - Works well with ESP32 over HTTP or WebSocket.
- **AWS IoT / Google Cloud IoT:**
    - Enterprise-level platforms for scalable, secure IoT solutions.

Libraries & Tools:

- **Libraries for Sensors**:
    - Dallas Temperature.h, One Wire.h (for DS18B20)
    - DHT.h (if using DHT11/DHT22)
    - Adafruit_Sensor.h (general sensor framework)
- **Wi-Fi & Networking**:
    - WiFi.h or ESP8266WiFi.h

    - HTTPClient.h or PubSubClient (for MQTT)
- **Data Format**:
    - JSON or CSV via built-in ArduinoJson.h or custom formatting.

Other Tools:

- **Serial Monitor / Plotter** – For debugging and real-time visualization.
- **Logic Analyzer / Oscilloscope** – For low-level sensor or signal debugging (advanced).
- **Fritzing** – For drawing circuit diagrams.
- **Git** – Version control, especially for collaborative projects.

Installation and Maintenance Tools:

Specialized Installation Tools:

These include tools like PV crimping tools, solar panel lifts, cable tie guns, and roof anchoring systems.

Electrical Test Equipment:

This includes meters and testers for measuring voltage, current, insulation resistance, and other parameters.

Thermal Imaging Cameras:

These help identify potential issues like overheating or faulty connections.

Increased Efficiency: Sensors can automate processes, optimize resource utilization, and reduce waste.

Remote Monitoring: Sensors enable continuous monitoring of systems and environments from a distance, facilitating early detection of issues.

CHAPTER 6
SYSTEM IMPLEMENTATION

6.1. Dataset Description:

A solar system dataset could describe various aspects of solar power generation, including PV system data, energy production, and weather conditions. It might include data on PV module performance, energy generation, and weather data from domestic premises or substations with high solar PV uptake. Other datasets might focus on solar resource assessment, including meteorological data like sunlight, wind speed, and temperature.

Solar Power Generation Data:

PV System Data:

This could include information on PV module types, sizes, and configurations, as well as sensor readings from inverters and plant-level sensors.

Energy Generation Data:

This dataset could contain data on the amount of electricity generated by solar panels over time, including hourly, daily, or weekly data.

Weather Data:

Datasets might include weather conditions like solar radiation, temperature, wind speed, and cloud cover, which significantly impact PV generation.

Solar Resource Assessment Data:

Meteorological Data:

This data would include information on sunlight, wind speed, temperature, and other relevant factors for solar energy production.

Geographic Information:

This could include location coordinates, solar radiation maps, and other geographical data relevant to solar resource assessment.

Solar PV Installation Data:

Location Data:

Datasets could contain the geographic coordinates of solar PV installations, allowing for mapping and analysis of solar energy deployment.

Image Data:

Sentinel-2 satellite imagery or other image data can be used to identify and classify solar PV installations, as described in a Nature article.

Semantic Segmentation Data:

Datasets might include pixel-wise labels for solar PV installations, enabling the training of AI models for identifying and segmenting solar farms.

6.2. Modules description:

- power plant configuration and control
- solar radiation models
- weather data assimilation
- Typical Meteorological Year (TMY)

Power Plant Configuration and Control :

Large-scale grid-tied PV power plants could achieve high degrees of penetration and improved performances, thus avoiding much of their intermittent and uncontrollable nature, thanks to the integration of ES units.

The main goal with such a configuration is to be able to release the PV production from its current real-time weather dependence and enable future PV power plants to provide a controlled constant production, which could be traded on electricity markets.

Therefore, since the solar resource presents a highly stochastic behavior, the PV instantaneous production should be completed at each moment by the power exchange performed by the ESS in order to achieve the instantaneous value of power that the power plant has committed with the market.

Solar radiation models:

- Solar radiation models estimate the amount of solar radiation reaching the earth's surface. These models can be parametric, decomposition, or empirical.
- Use meteorological parameters like atmospheric turbidity, cloud cover, and water content to calculate solar radiation
- Require detailed information about atmospheric conditions

- Use correlations between global, direct, and diffuse components of solar radiation
- Easier to use once the Global Horizontal Irradiance (GHI) is known
- Pyranometers measure radiation within a hemispherical field of view

Weather data assimilation:

data assimilation is most widely known as a method for combining observations of meteorological variables such as temperature and atmospheric pressure with prior forecasts in order to initialize numerical forecast models.

Data assimilation combines recent observations with a previous weather forecast to obtain our best estimate of current atmospheric conditions.

Data assimilation was first used in the 1960s in numerical weather forecasting models, with the goal of providing short-term predictions of meteorological .

Data assimilation schemes generate initial conditions for forecasts by using observations to correct short-range forecasts. The corrected states provide initial.

Typical Meteorological Year (TMY):

A dataset containing a full year of hourly weather data for a specific location, compiled by selecting the most representative months from a much longer period of recorded weather data, essentially creating a "typical" year for that location, often used for simulating solar energy systems and building energy performance analysis.

Each month in the TMY is chosen from a historical weather record, picking the month that best represents the average conditions for that particular month.

TMY is primarily used in simulations to evaluate the performance of solar energy systems, building designs, and other climate-dependent technologies, as it allows for a standardized comparison across different locations.

6.3. Performance Analysis:

Performance analysis of a solar PV system involves evaluating its efficiency and energy production in comparison to theoretical maximum output. Key performance indicators include efficiency, capacity factor, performance ratio, and electrical output. This analysis helps identify losses, assess the impact of environmental factors, and optimize system design and operation.

Key Aspects of Performance Analysis:

Performance Ratio (PR):

This metric compares the actual energy output of a solar plant to its theoretical maximum output, providing a measure of overall system performance.

Capacity Factor:

It represents the ratio of the actual energy production of a solar PV system to its theoretical maximum energy production over a specific period, usually a year.

Efficiency:

The efficiency of a solar panel refers to its ability to convert sunlight into electricity.

Electrical Output:

This refers to the actual amount of electricity generated by the solar system.

Global Solar Radiation (GSR) data:

Essential for system design and performance assessment, including predicting energy output and evaluating the impact of shading.

Degradation Analysis:

Important for understanding how solar PV systems degrade over time due to environmental and operational factors, helping to maintain optimal performance.

Economic Analysis:

Assessing the cost-effectiveness of solar PV systems compared to traditional energy sources.

6.4. Performance Measures:

Key performance measures for a solar photovoltaic (PV) system include energy production, performance ratio, capacity factor, and efficiency. These measures help assess the system's efficiency, identify potential issues, and optimize performance.

Energy Production:

This refers to the total amount of electricity generated by the solar PV system over a specific period (e.g., a day, month, or year). It's typically measured in kilowatt-hours (kWh).

Performance Ratio (PR):

The PR compares the actual energy output of the system to its theoretical maximum output under ideal conditions. It's a key indicator of system efficiency, with values closer to 100% indicating higher efficiency.

Capacity Factor:

This metric measures the actual energy output of a solar PV system relative to its theoretical maximum output if it ran continuously at its full capacity. It's expressed as a percentage.

Efficiency:

This refers to the percentage of sunlight converted into electricity by the solar panels. While solar panels have varying efficiencies, modern panels can achieve efficiencies of up to 22.8%.

Specific Yield:

This metric quantifies the energy produced per unit of solar irradiance.

Capacity Utilization Factor (CUF):

Similar to capacity factor, CUF measures the percentage of time a solar PV plant is actually generating electricity.

Solar Insolation:

This refers to the amount of solar radiation received by a surface, which is a crucial factor affecting solar PV system performance.

Temperature:

Panel temperature can influence their efficiency, with excessively high or low temperatures potentially impacting performance.

Energy Development:

This refers to the overall process of developing and deploying solar energy projects, including factors like investment and project costs.

In a solar power system, the input is sunlight, which is converted into direct current (DC) electricity by solar panels. This DC electricity is then converted to alternating current (AC) electricity by inverters, which is the usable form for homes and businesses. The primary output is clean, renewable electricity.

6.5. Input and Output:

Input:

Sunlight is the primary input for solar power systems.

Process:

Solar panels absorb sunlight and convert it into DC electricity.

Inverters convert the DC electricity to AC electricity.

Output:

The main output is clean and renewable AC electricity.

The amount of electricity produced depends on various factors, including sunlight intensity, panel efficiency, and shading.

Additional Factors:

The amount of electricity produced can be calculated by considering the wattage of the panels and the hours of sunlight exposure.

Inverters play a crucial role in converting the DC electricity from solar panels to the AC electricity needed for use in homes and businesses.

6.6. Results and Discussions:

Solar panels effectively convert sunlight into electricity, offering a renewable and environmentally friendly energy source. Their use is widespread, with applications ranging from residential rooftops to large-scale solar farms. While offering numerous benefits like reduced energy costs and environmental impact, solar panels also face challenges such as initial costs, reliance on sunlight, and potential for hot spots.

Results:

Energy Generation:

Solar panels effectively convert sunlight into electricity, providing a clean and renewable energy source.

Cost Savings:

Homeowners and businesses can reduce their reliance on traditional energy grids, leading to lower electricity bills.

Environmental Benefits:

Solar panels produce no greenhouse gas emissions during operation and can help reduce overall emissions when replacing fossil fuels.

Efficiency:

Solar panel efficiency has been steadily improving, with record-breaking efficiencies achieved in recent years.

Deployment Growth:

Global solar PV capacity is growing at a rapid pace, driven by cost reductions and supportive policies.

Material Recovery:

Research is ongoing to improve the recycling of materials from end-of-life solar panels.

Discussion:

Efficiency vs Cost:

While efficiency is constantly improving, the initial cost of solar panels and installation can be a barrier to entry.

Reliance on Sunlight:

Solar panel performance is dependent on the availability of sunlight, which can be affected by weather conditions and geographic location.

Hot Spots:

Malfunctions in solar panels can lead to hot spots, which reduce efficiency and potentially damage the panels.

Environmental Impact:

While solar panels offer significant environmental benefits, their production and end-of-life disposal also have environmental impacts.

Future of Solar:

Solar energy is projected to play a major role in the future of renewable energy, with continued advancements in technology and deployment.

India's Solar Energy Landscape:

Rapid Growth:

India's solar power sector is one of the fastest-growing globally, with ambitious targets and policies in place to support its expansion.

Government Initiatives:

The Indian government is actively promoting solar energy through various schemes and policies, including the National Solar Mission.

Hybrid Systems:

Hybrid solar-wind systems are being increasingly used to harness the power of both renewable resources, further enhancing reliability and efficiency.

Hot Spots:

Malfunctions in solar panels can lead to hot spots, which reduce efficiency and potentially damage the panels.

Environmental Impact:

While solar panels offer significant environmental benefits, their production and end-of-life disposal also have environmental impacts.

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Similar to capacity factor, CUF measures the percentage of time a solar PV plant is actually generating electricity.

Solar Insolation:

This refers to the amount of solar radiation received by a surface, which is a crucial factor affecting solar PV system performance.

Temperature:

Panel temperature can influence their efficiency, with excessively high or low temperatures potentially impacting performance.

CHAPTER 7

CONCLUSION AND FUTURE SCOPE

7.1 CONCLUSION

A solar power system is a reliable, eco-friendly solution that harnesses energy from the sun to produce electricity. By converting sunlight into usable power through photovoltaic technology, solar systems help reduce dependence on fossil fuels, lower electricity bills, and minimize environmental impact. Advancements in technology such as efficient solar panels, battery storage, inverters, and smart monitoring software have made solar energy systems more accessible, efficient, and scalable for both residential and commercial applications.

While there are initial costs and system design considerations, the long-term benefits in terms of cost savings, energy independence, and sustainability make solar power a smart investment for the future. As the world shifts toward renewable energy, solar systems will play a vital role in achieving clean energy goals and building a more sustainable planet.

7.2 FUTURE ENHANCEMENT

The future of solar energy is bright, with rapid advancements driving higher efficiency, smarter systems, and wider adoption. Here are key areas of future enhancement in solar energy.
Future enhancements of solar panels will focus on improving efficiency, reducing costs, and increasing versatility.This includes developing new materials, enhancing manufacturing processes, and integrating solar panels into more applications.

Future advancements in solar panel technology focus on improved efficiency, enhanced durability, and greater integration with the built environment and energy storage systems.Key areas of development include material innovations like perovskite and advanced silicon, as well as hybrid panel designs combining different materials for optimal efficiency. Additionally, smart solar systems with integrated management and energy storage are gaining traction.

APPENDIX 1
SOURCE CODE

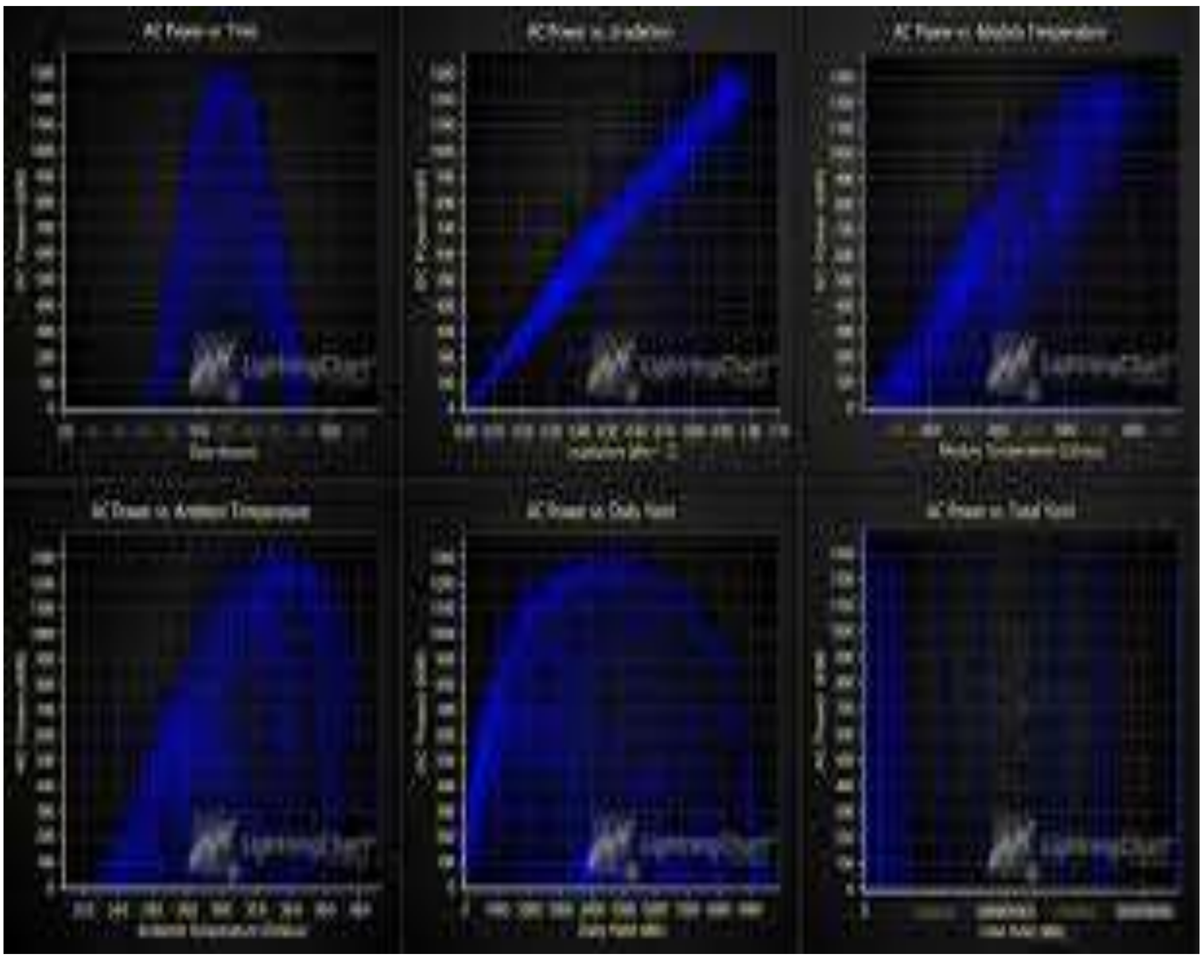
Package:

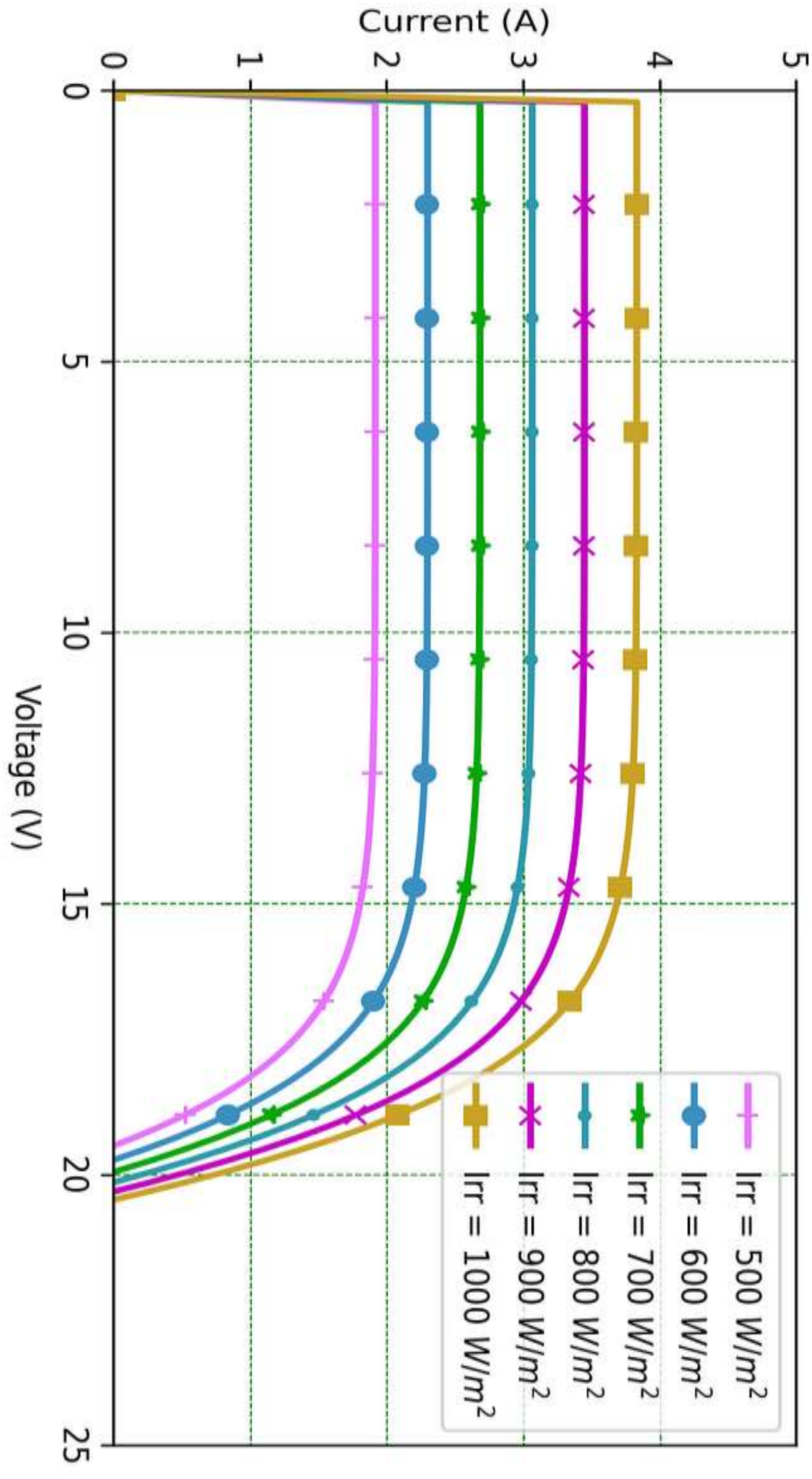
```
#include <ESP8266WiFi.h>
#include <OneWire.h>
#include <DallasTemperature.h>
#include <ThingSpeak.h>
const char* ssid = "YOUR_WIFI_SSID";
const char* password = "YOUR_WIFI_PASSWORD";
unsigned long myChannelNumber =
YOUR_CHANNEL_NUMBER;
const char* myWriteAPIKey = "YOUR_API_KEY";
WiFiClient client;
 #define ONE_WIRE_BUS D2
 OneWire oneWire(ONE_WIRE_BUS);
 DallasTemperature sensors(&oneWire);
void setup()
{
 Serial.begin(115200);
 delay(100);
 sensors.begin();

 WiFi.begin(ssid, password);
 Serial.print("Connecting to WiFi...");
 while (WiFi.status() != WL_CONNECTED) {
  delay(500);
  Serial.print(".");
 }
 Serial.println(" connected!");
```

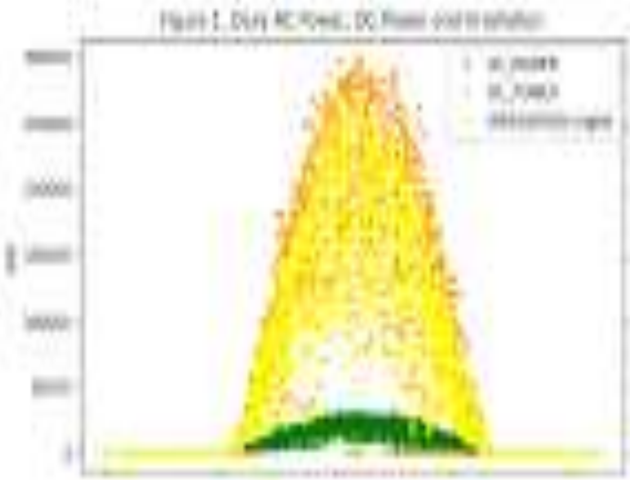
```
  ThingSpeak.begin(client);
}
void loop()
{
  sensors.requestTemperatures();
  float temperatureC = sensors.getTempCByIndex(0);
  Serial.print("Temperature: ");
  Serial.print(temperatureC);
  Serial.println(" °C");
  ThingSpeak.setField(1, temperatureC);
  int statusCode = ThingSpeak.writeFields(myChannelNumber,
myWriteAPIKey);
  if (statusCode == 200)
{
    Serial.println("Data sent to ThingSpeak successfully.");
  }
  else
{
    Serial.print("Error sending data: ");
    Serial.println(statusCode);
  }
     delay(60000);
}
```

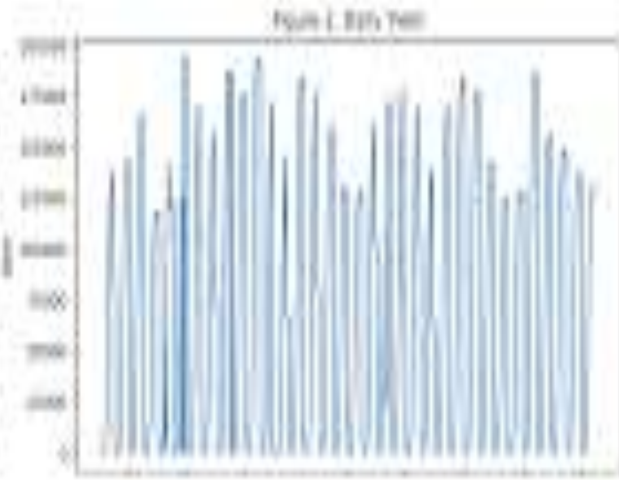
APPENDIX 2

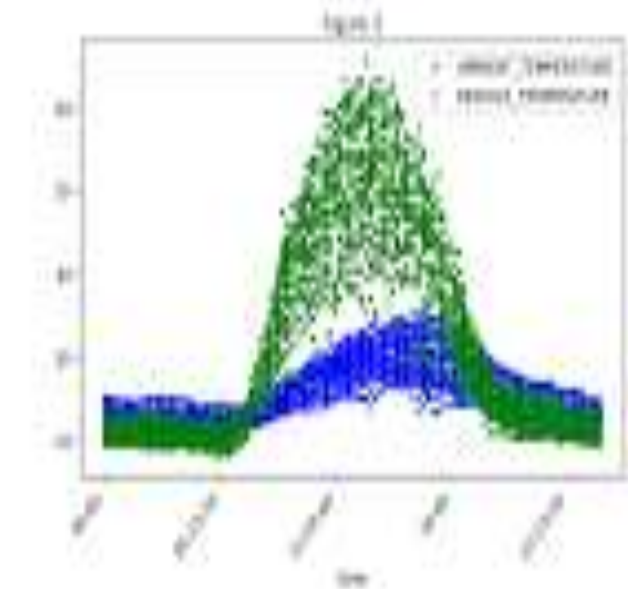
SCREENSHOTS :

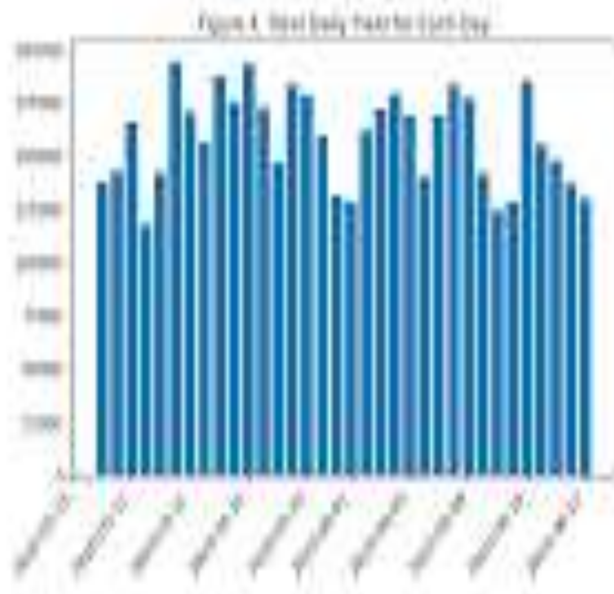



Current (A)
0
1
2
3
4
5
Voltage (V)
0
5
10
15
20
25
Irr = 500 W/m^2
Irr = 600 W/m^2
Irr = 700 W/m^2
Irr = 800 W/m^2
Irr = 900 W/m^2
Irr = 1000 W/m^2









APPENDIX 3

LIST OF PUBLICATION

Mr.M.DEEPAK, Mr.S.ANANDHAKUMAR, Mr.V.BHUVANESHWARAN, Mr.R.ARULSURIYAN, **"SOLAR ENERGY ESTIMATION FOR MINIMIZING ENERGY STORAGE REQUIREMENTS IN PV POWER PLANTS"**, Paper has been accepted for the AICTE sponsored International Conference On Communication, Electronics and Electrical Engineering (ICCEEE2024).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Certificate
of Participation

This is to certify that Dr./Prof./Mr./Ms. ANANDHAKUMAR . S - IT

of GNANAMANI COLLEGE OF TECHNOLOGY

has participated and presented a paper entitled SOLAR ENERGY ESTIMATION FOR MINIMIZING ENERGY STORAGE REQUIREMENT IN PV SYSTEM

in the **National Level Conference on Innovations in Information, Computing, and Artificial Intelligence Technology (NCIICAIT 2K25)** organized by the Department of AI&DS, CSE and IT on 07th April, 2025.

Convenor

Principal

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PROGRAM OUTCOMES (POs)

PO 1	Engineering knowledge	Apply the knowledge of mathematics, science, engineering fundamentals and an engineering specialization to the solution of complex engineering problems.
PO 2	Problem analysis	Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO 3	Design/development of solutions	Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO 4	Conduct investigations of complex problems	Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO 5	Modern tool usage	Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO 6	The engineer and society	Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO 7	Environment and sustainability	Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO 8	Ethics	Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO 9	Individual and team work	Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO 10	Communication	Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO 11	Project management and finance	Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO 12	Life-long learning	Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Course Code & Name :C407-(IT3811-PROJECT)
REGULATION :2021
YEAR/ SEM :IV/VIII

COURSE OUTCOMES

C412.1	G	Gain domain knowledge and technical skill set required for solving industry / research problems.
C412.2	P	Provide solution architecture, module level designs, algorithms.
C412.3	I	Implement, test and deploy the solution for the target platform.
C412.4	P	Prepare detailed technical report, demonstrate and present the work.

CO/PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PO12	PSO 1	PSO 2
C412.1	3	2	2	2	2	2	-	2	3	2	2	3	3	3
C412.2	3	3	2	3	3	3	-	3	3	3	2	3	3	3
C412.3	3	3	3	3	3	2	2	3	3	3	2	3	2	3
C412.4	3	3	3	3	3	2	3	3	3	3	3	3	3	3
C412.5	3	3	3	3	3	2	3	3	3	3	3	3	3	3
C412	3.00	2.80	2.60	2.80	2.80	2.20	2.67	2.80	3.00	2.80	2.40	3.00	2.80	3.00